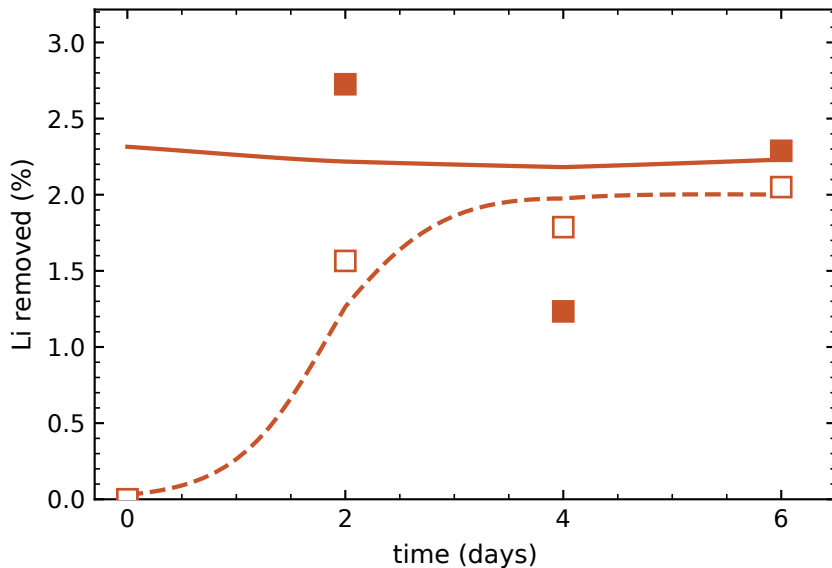
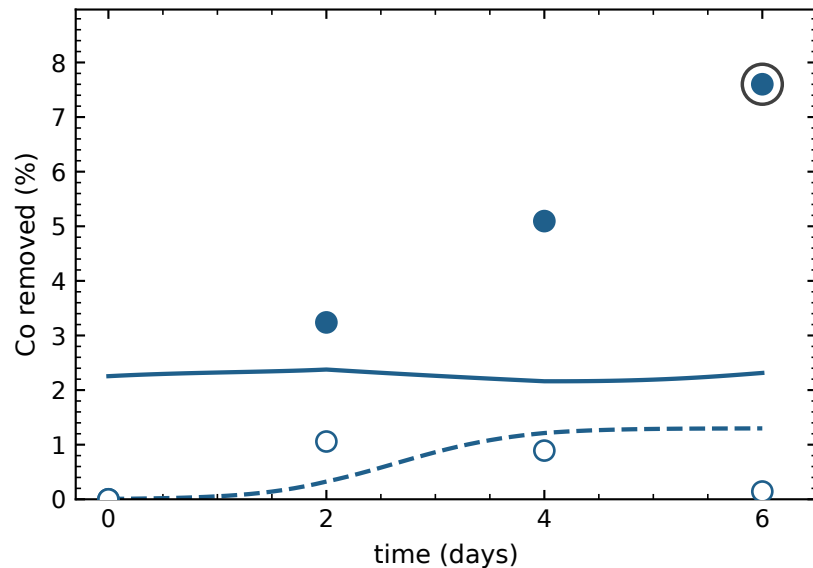


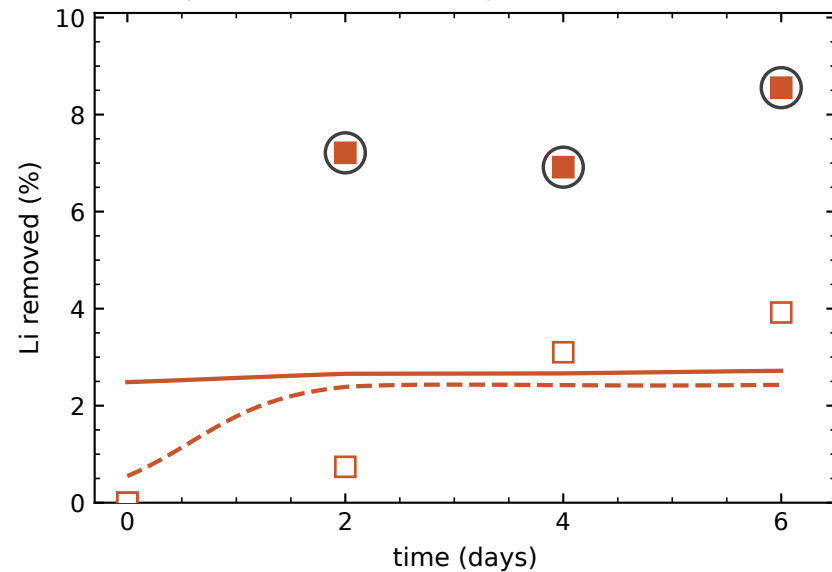
(A) Li, single ion — $\log K_{int} = -2.39$



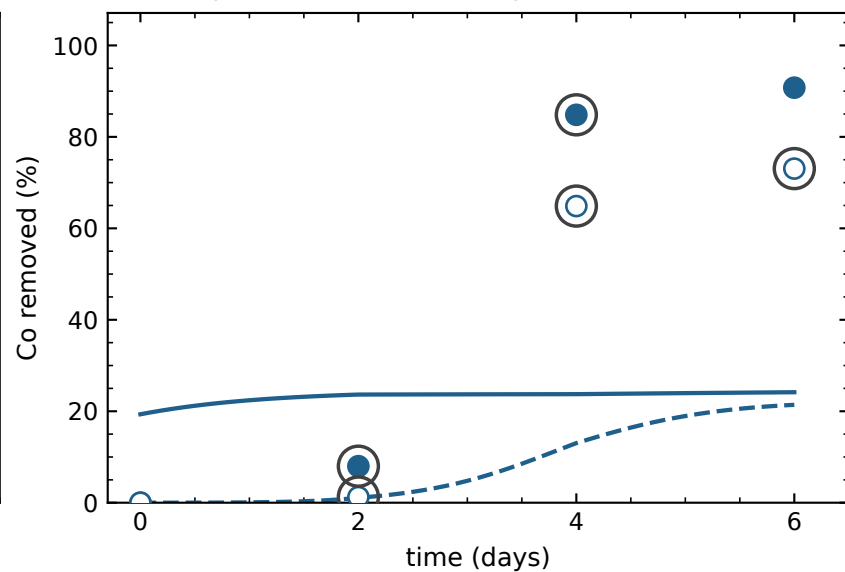
(B) Co, single ion — $\log K_{int} = -2.34$



(C) Li, produced water — $\log K_{int} = -1.13$



(D) Co, produced water — $\log K_{int} = +0.65$



■ measured, pH 6 start □ measured, pH 2 start — model, pH 6 start - - model, pH 2 start ○ excluded: carbonate supersaturated or above one monolayer